

Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix

Source: Our World in Data – Renewable Energy

75% of Global GHG Emissions Still Come from Fossil Fuels

75%

of global greenhouse gas
emissions come from
burning
fossil fuels for energy

5 Million+

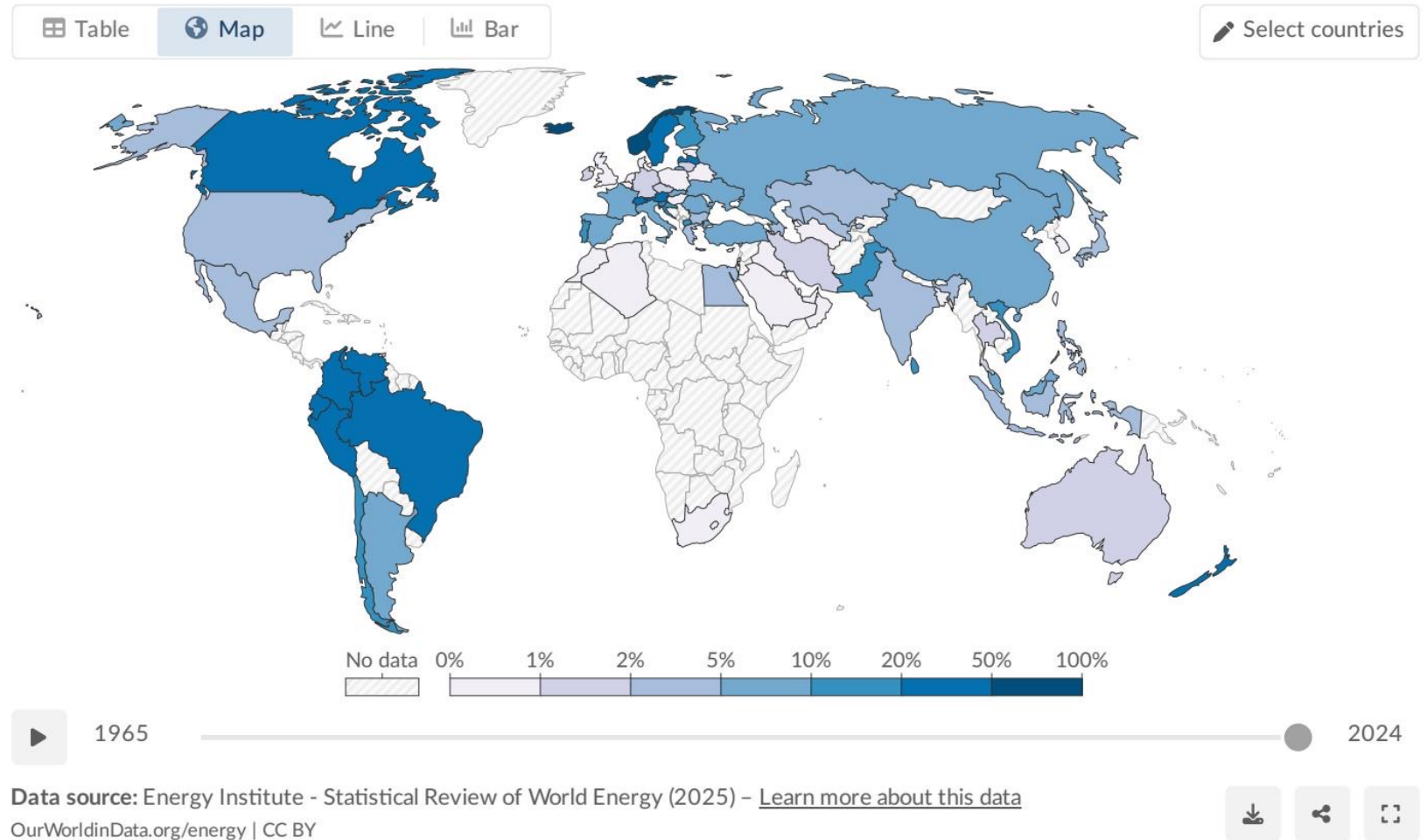
premature deaths per year
caused by local air pollution
from fossil fuel combustion

Urgent Shift

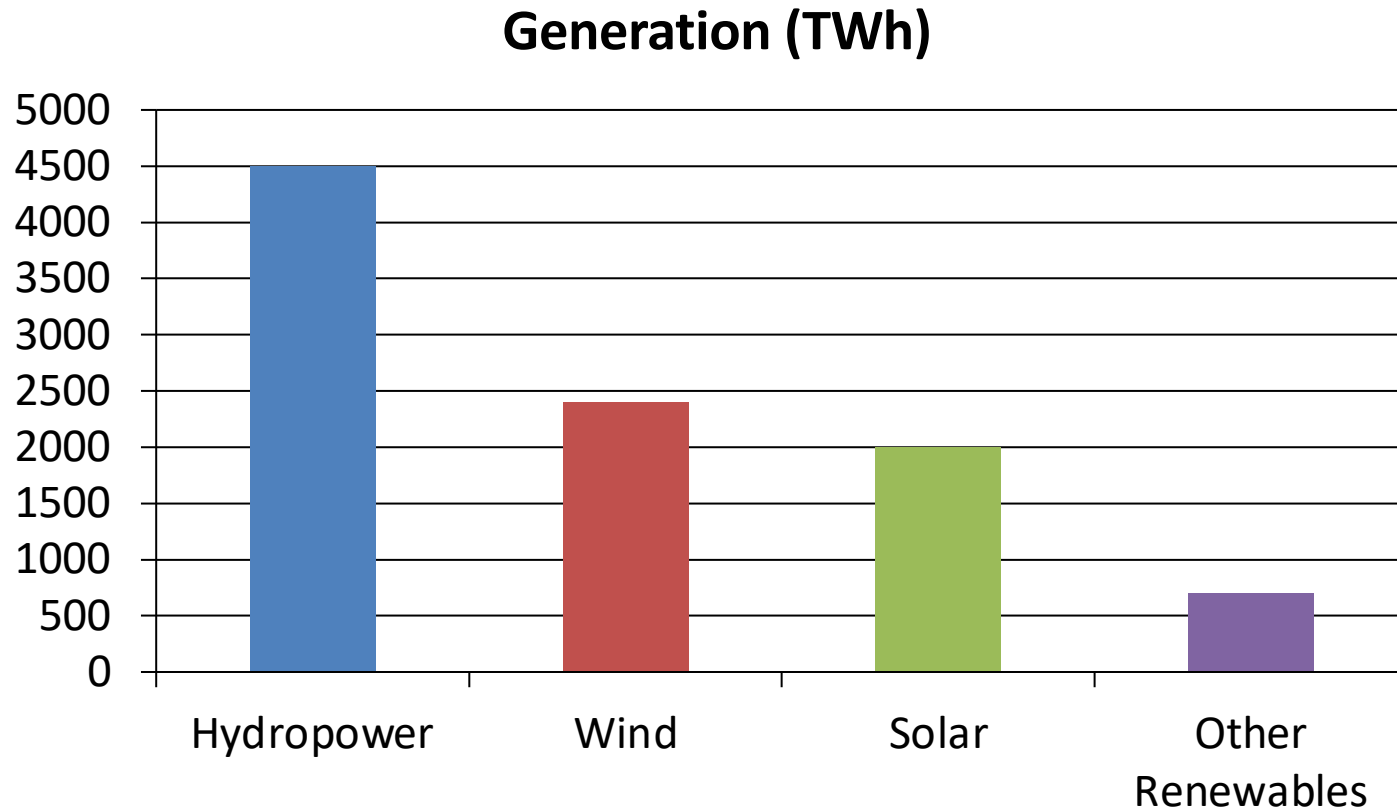
The world must rapidly shift
toward low-carbon energy
sources:
nuclear and renewables

Renewables Now Supply ~14% of Global Primary Energy

- Approximately one-seventh (~14%) of the world's primary energy now comes from renewable technologies (2024 data, substitution method).
- Primary energy encompasses electricity, transport, and heating combined.
- Because transport and heating are harder to decarbonize, the renewable share in the total energy mix is lower than in electricity alone.
- Regional variation is significant: Norway, Brazil, and New Zealand



Renewable Electricity Generation Is Growing Rapidly, Led by Wind and



- Wind and solar generation has accelerated since ~2010.
- Wind and solar combined (4,400 TWh) now nearly match hydropower's output (4,500 TWh).
- Other renewables include geothermal, biomass, wave, and tidal sources.

Almost One-Third of Global Electricity Now Comes from Renewables

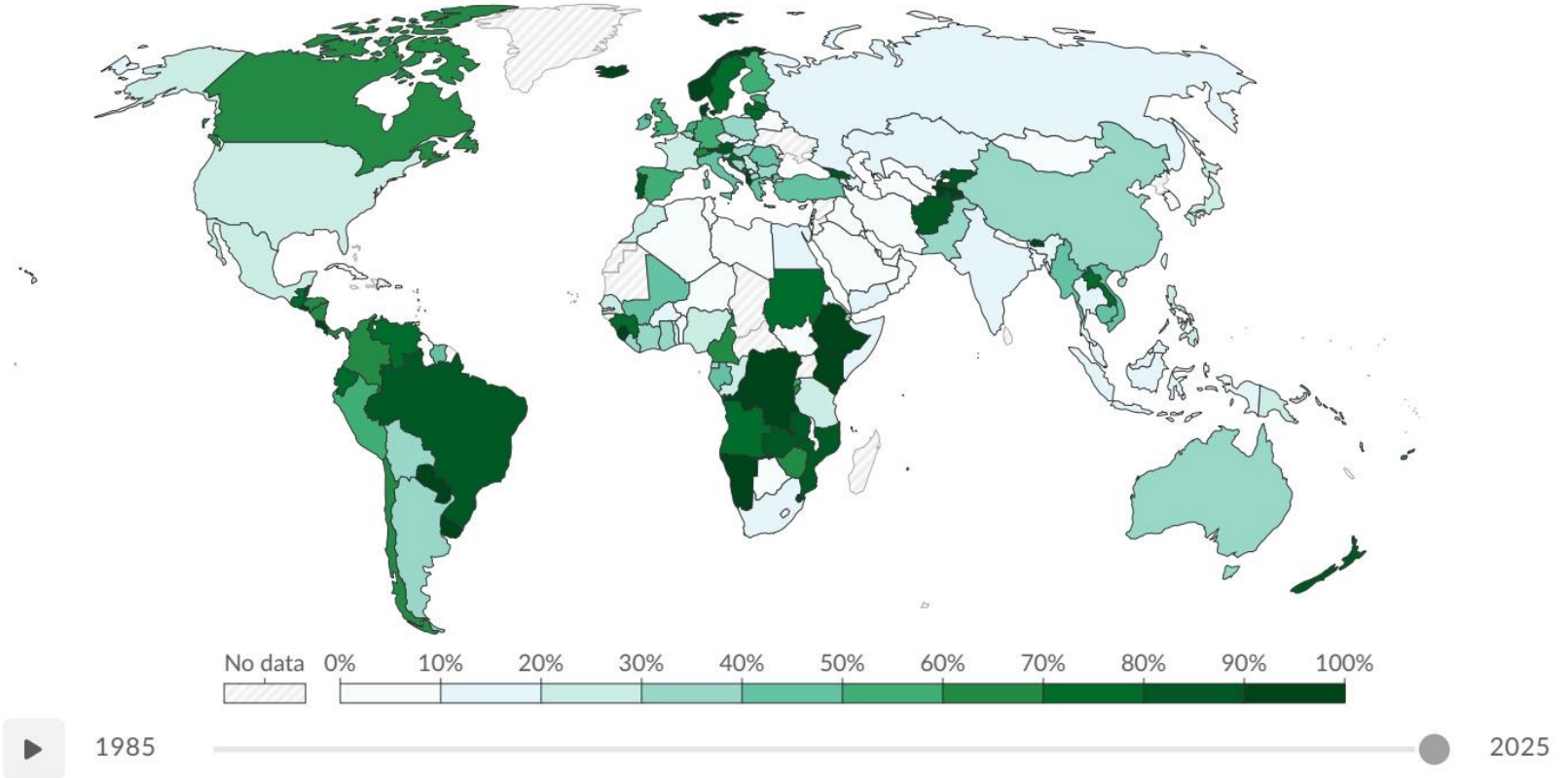
Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

Our World
in Data

Table Map Line Bar

Select countries

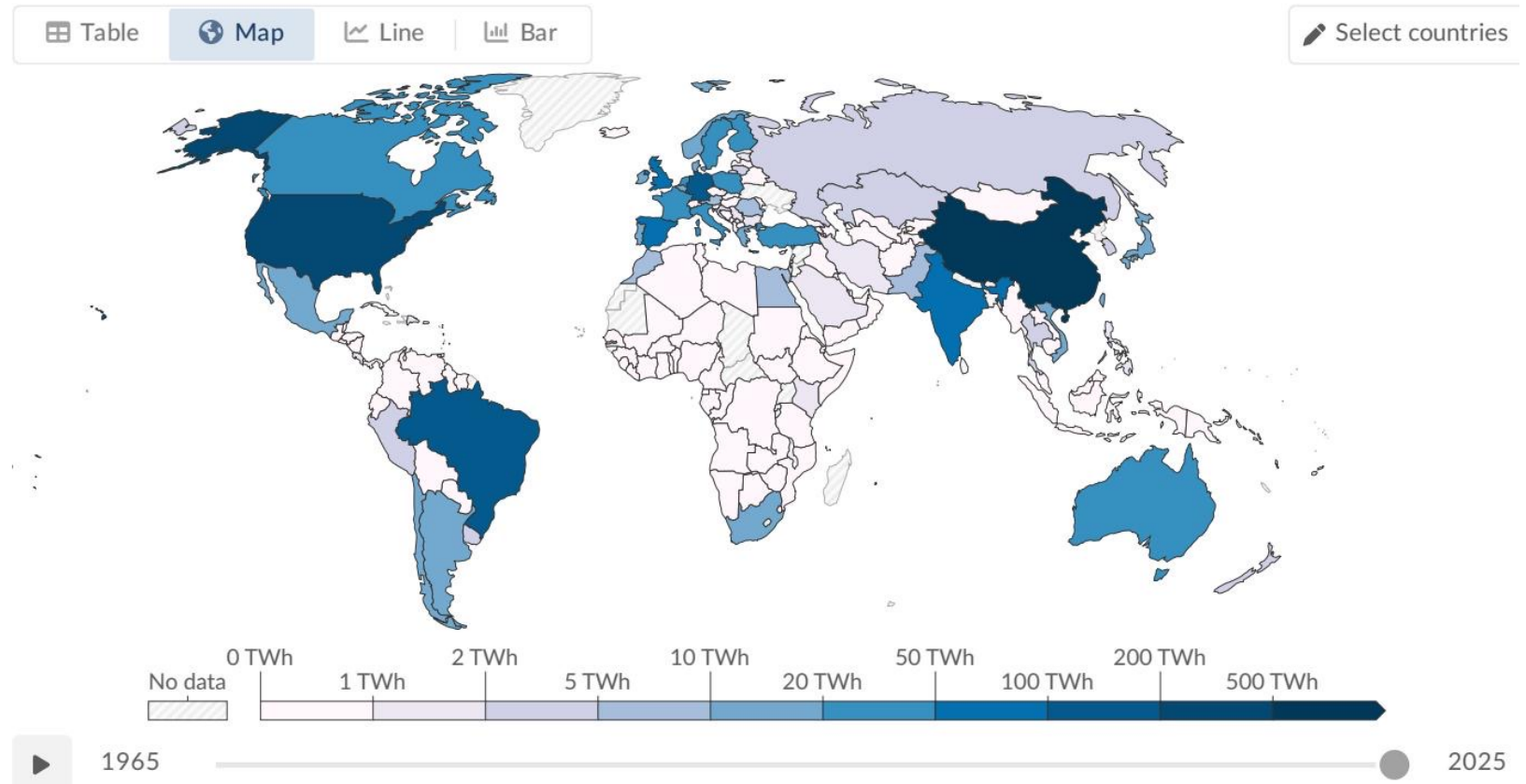


Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) - [Learn more about this data](#)
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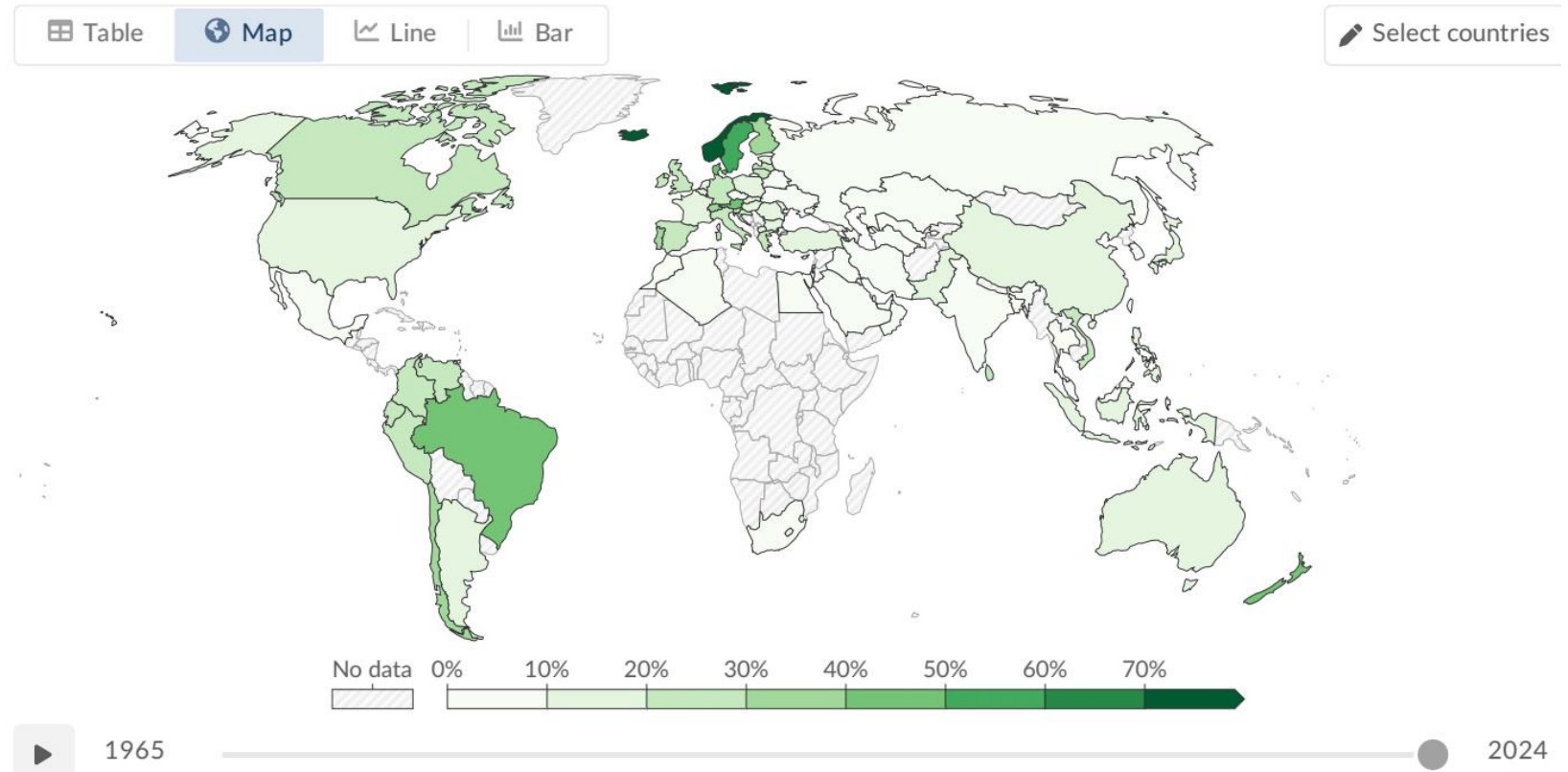
Hydropower Remains the Largest Renewable Source at ~50% of Gene

- Hydropower is the single largest renewable electricity source.
- Accounts for roughly half of all renewable electricity (~47%).
- It has been generating electricity at scale for over a century.
- Top producers include China, Brazil, and Canada.



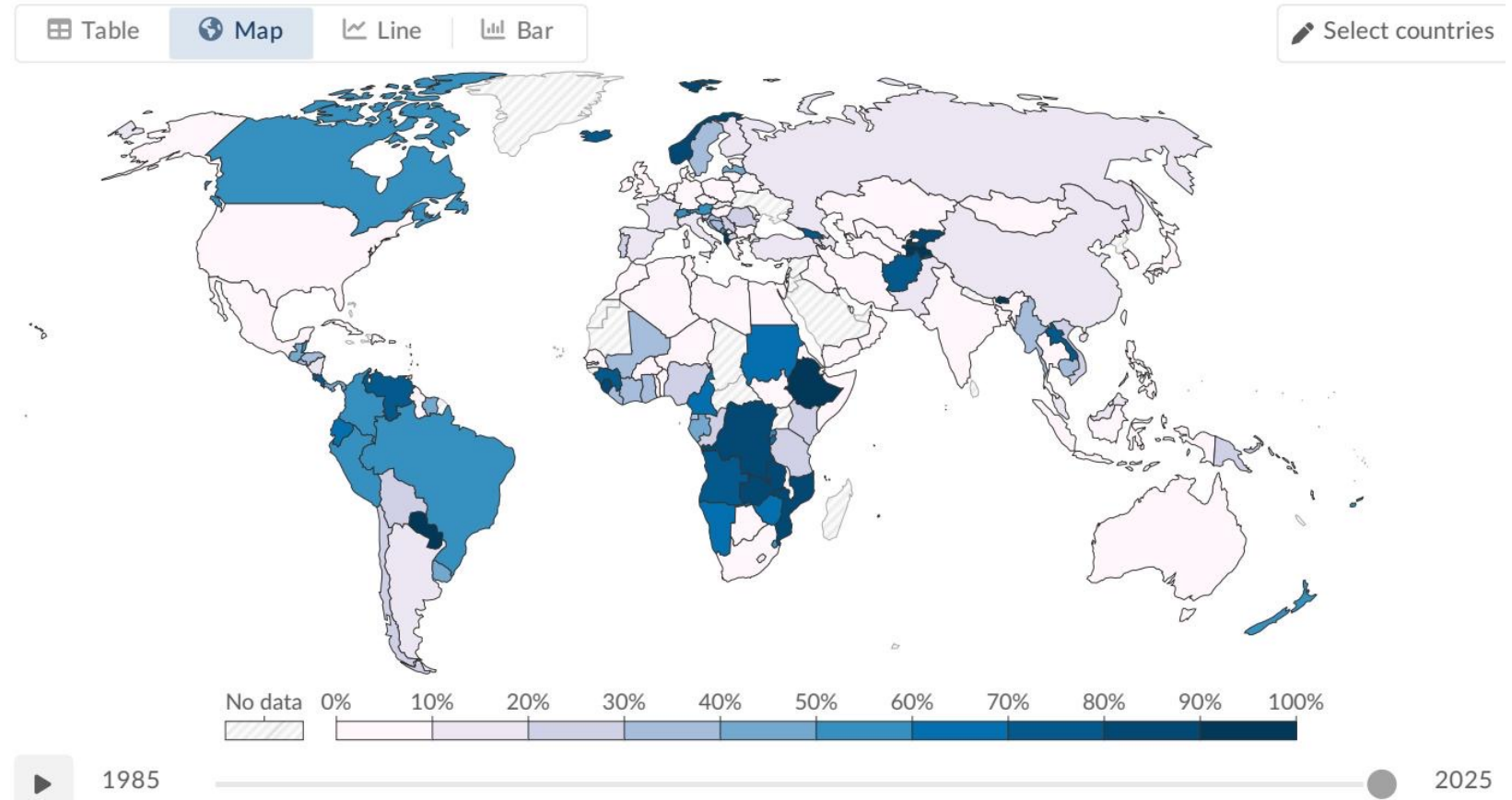
Wind and Solar Are the Fastest-Growing Energy Technologies in History

- Wind: ~2,400 TWh (~25% of renewable electricity). Major growth in Europe, China, and US.
- Solar: ~2,000 TWh (~21%). The fastest-growing source, driven by falling costs and policy support.
- Combined, they now nearly match hydropower output.
- Both technologies have seen dramatic acceleration since 2010.



Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and L

- Scandinavia, Brazil, and Canada lead with 65–98% renewable electricity shares, benefiting from abundant hydropower.
- Denmark and Germany have invested heavily in wind and solar.
- Many African and Middle Eastern countries have renewable electricity shares below 10%.
- Sub-Saharan Africa has minimal renewable infrastructure.
- Middle Eastern nations remain almost entirely fossil-fuel dependent.



Key Takeaways: Renewables Are Growing Fast but the Transition Must

- Milestone reached: ~14% of global primary energy and ~30% of global electricity now come from renewable sources.
- Hydropower remains dominant (generating roughly half of all renewable electricity) but its growth potential is limited compared to wind and solar.
- Wind and solar are the fastest-growing energy technologies in history; combined, they now produce nearly as much electricity as hydropower.
- Stark regional disparities persist: Scandinavia and Latin America exceed 60–90% renewable electricity, while much of the Middle East and Sub-Saharan Africa remain below 10%.
- Transport and heating lag far behind electricity in the renewable transition.
- Forward Outlook: The pace of transition must accelerate across all sectors to meet